

Financial Markets Development: Evidence for Brazil, India and China

Abstract

The article analyses the level of financial markets development in a sample of 42 countries and compare such level with economic, international integration and cultural characteristics of these countries. The results show that greater international integration, lower inflation, greater focus on entrepreneurship and greater press freedom, among others, are characteristics which exist in countries with more developed financial markets. We then analyze the Brazilian, Indian and Chinese characteristics *versus* the characteristics for the entire sample.

Keywords: BRICs, financial markets, development

I – Introduction

There is no doubt that developed financial markets are the basis for economic growth. Since Goldsmith (1969) such fact has been very well covered in the literature. However, not only the level of financial markets development is important, but also its architecture – which is the greater orientation of the economy toward a banking or market system. As examples of such orientation, we can see United States and England, whose economies are oriented to stock markets, and Germany and Japan, whose economies are oriented to banking system. Several authors, such as mentioned by Dermigue-Kunt and Levine (1999), indicate advantages and disadvantages in both systems. According to those who support banking systems, they play a positive role in identifying adequate projects, mobilizing the necessary capital, monitoring managers and risks. Besides, in countries whose legal structure is incipient, powerful banks may force companies to disclose information and pay their debts. Those who support market systems point the disadvantages of bank-controlled economies; the first one of them is the bias for low-risk projects, what retards economic development. On the other hand, according to these very same supporters, the advantages of market system are more disclosure of information, more easiness to diversify investments and standardization of risk-management instruments.

Given the importance of financial markets for economic growth, a natural questioning may be raised: which are the characteristics that distinguish countries with different levels of financial market development? I.e., which are the characteristics of countries whose financial market is more developed or less developed?

In the late 20th century, a new academic approach was raised to try to explain the development of different markets. Instead of the neoclassic analysis based on decreasing

capital earnings, we started to deal with institutional characteristics in the model, mainly with the cultural / legal system of the different countries. Several authors, among them La Porta, dedicated themselves to this kind of study, dividing the countries according to the origin of their cultural / legal system between common law and civil (or code) law; or, as an alternative, between systems deriving from the British tradition (common law) and from the European, French, German or Scandinavian tradition (code law). A small digression may help to understand this issue. On the one hand there are countries in which the legal codes were built and imposed in a certain moment or in relatively short periods. On the other hand there are countries in which legislation was born from uses and traditions. In the first group, one can see Scandinavian countries that organized their legal codes in the 17th and 18th centuries, and those codes have remained unaltered and immune to German or French influences throughout the centuries. In France, Napoleon organized the French Civil Code and obliged its adoption in conquered countries such as Italy, Poland, Netherland and the Austro-Hungarian Empire. The influence of such code extended to Indochina, North and Sub-Saharan Africa, Oceania and French Guiana throughout the years. Besides, it was the basis to the development of the legal system in Spain and Portugal, thus influencing also Latin American, including Brazil. The German Code, finished in the end of the 19th century, exerted influence on Austria, Switzerland, China, Taiwan, Czech Republic, Slovakia, Greece, Hungary, Italy, countries that formed the former Republic of Yugoslavia, Japan and Korea.

Contrariwise to the elaboration of a strict code of conduct, the so-called civil law (or English) system has laws that originate from the individual solution of suits by the judges. This system exerted influence on Caribbean, North America and parts of Asia, such as India and Africa.

Besides the original kind of development, the two systems may be differentiated by the role of the judges. In the common law system, they have freedom to act and decide each suit according to the circumstances, whereas in the code law system there is no such freedom, and the decisions are directly associated to the legislation. This difference makes the common law system be more adaptive and quicker to follow the evolution of customs. Consequently, it favors the financial markets development. Another view about this process shows that the code law system protects the State, whereas the common law system protects private property.

Those who support the absolute influence of the legal system on the level of market development, such as La Porta *et al* (1998), argue that the European systems are very strict, what impedes the formation of a entrepreneurs class and, at the same time, does not adequately protect the investors.

After some years in which the connection between the origin of the legal system and the market development totally prevailed, other views have recently been raised, such as Modigliani and Perotti (1997), in favor of the list of determinants of financial markets development. Aspects more related to the quotidian of economic agents, for instance protection to investors, role of the legal system and level of corruption, started to be taken into account in the analysis. And that happens due to the discovery, by these authors, that there is no exact dichotomy between economies oriented to banks or to financial markets. In fact, at least in part of the countries, both systems exist together and the best one can do to analyze them is identifying metrics which may consider the existence of both systems, instead of simply separating the countries in a pure binomial universe.

In this work we will analyze the existence of characteristics which may at least partially explain the differences in the level of financial markets development around the world, specially the cases of Brazil, India and China, which have been the preferred target of foreign investors. First of all, the main studies already carried in this field will be described and then we will explain our analyses. Section II is the literature review on this subject, section III presents the used data, section IV describes the methodology and results and, finally, section V presents the main conclusions.

II – Literature Review

La Porta, Lopez-de-Silanes, Schleifer and Vishny (1998 and 1999) come to the conclusion that countries whose legal system is based on code law have lower level of economic development than the others. Such countries in particular present lower level of public companies' market value on GDP – Gross Domestic Product and also a smaller number of Initial Public Offerings. The authors also show that countries based on code Law offer less rights to stockholders. Besides, such countries present less exigencies regarding disclosure of information. On the other hand, countries based on common law offer more protection to general creditors. It is important to notice that more protection to creditors leads to a greater development of financial intermediation, more protection to stockholders and greater development of the stock market. One can infer that countries with common law system will have more developed both credit and stock markets.

To corroborate such hypothesis, Dermigue-Kunt and Levine (1999) analyze the financial structure of 150 countries regarding three aspects: economic development *vis-a-vis*

the bank and stock market development, economic development and systems based on market *versus* systems based on banks and, finally, the determinants of the financial structure, taking into account legal, regulatory, taxation and macroeconomic aspects. Their main conclusion is that countries whose legal system is based on common law and have strong protection to minority stockholders, good accounting standards, low level of corruption and no insurance to bank deposits tend to be more market-oriented. On the other hand, countries whose legal system are based on the French system (code law) and have low protection to minority stockholders, low guarantee of contracts, high level of corruption, bad accounting standard and high inflation level tend to have less developed financial systems, even when controlled for income level of the countries.

However, another point of view originated from this one and it argues that not only the origin of the cultural / legal system is a determinant to financial markets development, but also other institutional aspects, which matter a lot. Aspects such as protection to investors and stockholders, corruption, political risk and others have been the object of several studies. La Porta *et al* (2000), in particular, when completing their former studies, state that the efficiency of the legal system is the main provider of conditions that will determine innovation and economic growth. I.e., it is not the origin of the legal system that determines the characteristics of the economic system, but its efficiency.

Likewise, Beck and Levine (2003) describe the so-called Law and Finance Theory, which focuses on the role of the legal system of each country as an explanation to the differences between financial markets. The first part of the theory states that in countries where legal institutions support private property and contracts and protect the investors' rights, the savers are more willing to finance new projects and thus the financial markets bloom. Contrariwise, in countries where legal institutions do not give such support, there is more inhibition to financing new projects and low financial development. The second part of the theory affirms that different legal traditions, which appeared in Europe and spread around the world through colonization, conquering or imitation, explain part of the differences in financial development observed nowadays.

Tadesse (2002) relates the financial architecture of difference countries, i.e., the level in which the financial system is more oriented to banks or to the market, with economic development. The author supports the thesis that it is not only the origin of the legal system that will define the economic development, but a cluster of factors. He highlights an advantage of more market-oriented systems: the prices formed in it have important information content. The author uses architecture indicators regarding size, activity and

efficiency of the stock market in relation to the bank sector and aggregates such measures for the tests. Size is measured by the quotient between the volume of capitalization of companies listed in the stock Exchange by the volume of bank credit in the private sector. Activity is measured by the quotient between the total value of traded stocks by the volume of bank credits. Finally, efficiency is measured by the product of the total value of traded stocks, measured as percentage of GDP, and the indirect costs of the banking sector divided by the total assets of the same sector. The variable architecture will be the arithmetic mean of the means of these three indicators. Another measure used in this study regards the level of market development. The author considers as underdeveloped the countries that remain under the mean for two indicators: bank and market development. After analyzing data from 36 countries concerning financial architecture, financial development and industry performance, the author concludes that there is a positive correlation between economic performance and level of orientation to the market in countries whose financial systems are developed; whereas in countries whose financial systems are low developed, such correlation is negative.

Chinn and Ito (2005) analyze how much financial liberalization contributes to the stock markets development. The authors indicate several benefits of financial liberalization, such as the reduction of capital cost and of artificiality in determining the interest rate. They use three measures of market development: market capitalization, total value of traded stocks and working of stock market. An index was developed for market openness and liberalization based on four binary variables: existence of multiple exchange rates, existence of restrictions for transactions with currency, existence of restrictions for transaction with capital and, finally, existence of restrictions for using export revenues. They conclude that financial liberalization contributes to the development of the stock market, but only when there is development on the legal and institutional system of the country. Another interesting results is that the consumer goods and capital markets openness is a pre-condition for financial openness. Lastly, another result shows that the development of a strong bank sector is also a pre-condition to the stock market development.

Levine and Zervos (1996) analyze the empirical relation between stock market development and economic growth in different countries in the long term. By using measures such as market capitalization over the GDP, liquidity given by the relation between medium value of trades over the GDP, and also market turnover ratio, and, finally, international integration based on a multifactor model IAPM - International Arbitrage Pricing Model, which was developed by Korajczyk (1996), the authors build an aggregated measure of markets development by calculating the mean of such measures. They also estimate the depth

of the market through the index of the value of financial system debts over the GDP. Based on data from 24 countries for the period from 1976 to 1996, they show a positive and strong relation between stock market development and economic growth. Moreover, they demonstrate a strong connection between the determinants of stock market development and economic growth.

Another study indirectly related to the subject was carried by Fan and Titman (2003). The authors study capital structure and the alternatives of maturity for corporate debt in 39 countries. They find evidence that the legal system, common law or code law, and the taxation system affect directly the choice of capital structure. They also indicate the positive impact of the existence of mutual funds addressed to institutional investors on the development of banking systems and stock markets.

Alfaro *et al* (2004) examine the relation between foreign direct investment, financial markets and economic growth. With a sample of more than 50 countries, they conclude that non-developed financial markets restrict the benefits that derive from direct investment. The authors highlight that, although their definition of non-developed financial markets is associated to restrictions on short term capital circulation, the benefits of foreign direct investment flow, which is composed of long term resources, are as well reduced.

Still among studies regarding financial architecture and economic growth, Beck *et al* (2000) use three methodologies. Firstly, they analyze whether countries whose financial system is based on banks grow more than those whose financial system is based on market. Then they examine whether industries which are strongly dependent on foreign investment grow more in market-based or bank-based systems. Finally, they investigate whether companies in the different systems grow under the rate predicted by the generation of internal resources and short term financing in bank-based and market-based systems. The three methodologies present the same results. The first one shows that the financial structure is not an adequate mechanism to distinguish financial systems; the second one indicates that the financial structure does not explain economic growth or the development of different industries or even expansion of companies. Finally, the authors show that the results are inconsistent with the traditional view of separating economies based on markets or banks. The element which distinguishes economies is the existence of a legal system that effectively protects foreign investors.

Ndikumana (2001) investigates the impact of the financial system architecture on domestic investment. Based on a sample of 99 countries in the period from 1965 to 1997, he classifies the countries into four groups: underdeveloped countries with financial system

based on market or based on banks and developed countries with financial system based on market or based on banks. The author tries to answer two questions. The first one concerns the relation between financial markets development and level of domestic investment. The second question is discovering whether there is any relation between architecture of markets and level of domestic investment. The empirical evidence supports the thesis that there is a positive relation between financial markets development and level of domestic investment. Such fact suggests that a more developed financial market will offer more abundant capital, cheaper and more efficiently allocated. When it comes to the second question, the empirical evidence suggests that there is no impact of the financial system architecture on the level of domestic investment. Such result is inconsistent with the idea of superiority of one financial system architecture over the other. According to the author, the results are consistent with the idea that markets and banks are complementary, and does not exclude each other.

Francis, Khudara and Pereira (2001) analyze the differences between accounting systems of countries with common law and code law systems. They examine data of 1990 for 31 countries and test two hypotheses. The first one relates countries with code law with low level of protection to investors and, thus, accounting systems with little transparency. The second one relates countries with code law with small demand for audit. The empirical results support both hypotheses. Countries with common law system have more transparent accounting systems and more demand for audit. The authors also identified differences between countries with code law. Among them, those who have more accounting transparency have more demand for audit.

Fuchs-Schündeln and Funke (2001) analyze the effects of liberalization in stock markets on financial and economic development in 27 countries. The analyzed data concerns the period from 1975 to 2000, with data from countries which implemented liberalization programs between 1980 and 1995. The results indicate that liberalization is associated with an increase on private investments in the short term of 14% in the first four years since the liberalization and of 4% in per capita income for the same period. The growth tends to be greater if liberalization is preceded by institutional reforms. The authors also found evidence of a permanent growth effect of 0.4% based on a sample of 72 countries.

Boutchkova and Megginson (2000) study the impact of privatizations on stock markets. Based on a sample of 31 countries and data from 1983 to 1999, the authors verify a decrease in the participation of banks on financial markets of support to corporation around the world. The fraction of domestic credit withheld by banks in the countries of the sample remained stable – around 125% - during the analyzed period. On the other hand, the stock exchanges'

capitalization value rose from 52% to 82% of the GDP in the analyzed countries. The stocks that were issued in privatization programs significantly contributed to an increase of more than 11 times the total of markets capitalization as well as the great growth in the volume of trades in the stock exchanges, which rose from US\$ 1.23 trillion in 1983 to US\$ 37.5 trillion in 1999. The liquidity of the markets was also affected by privatizations. The authors indicate that it is responsible for an increase of 2.3% in the first year after the privatization and of 1.7% in the second one. Privatized companies are the most valuable on six out of the ten biggest stock markets of the world, excluding the USA. The practices of investment banks were changed, as 23 out of the 25 greatest issuances of stocks were of privatized companies. They also show empirical evidence favorable to privatization regarding the investors' point-of-view. Besides the excess return to investors, privatizations have brought a wide number of new investors to different markets. Almost two thirds of non-American companies which have more than 500,000 stockholders were born from privatization. More than a dozen of them have more than 1,000,000 stockholders. Nevertheless, the authors point out that this basis of stockholders is not stable, and it had a one third reduction in the first years after the privatization. The authors conclude that privatizations brought many benefits to stock markets.

Modigliani and Perroti (1997) use as a proxy to the quality of legal and juridical system the premium of stocks with voting right over those which do not have such right. In countries where such premium is great, the financing of companies is dominated by banks and the stock market is small. The authors show that where investors' rights are not well protected, there is difficulty for companies to issue stocks and, as a result, new lucrative ventures are not developed.

Herrero *et al* (2002) carried a descriptive analysis of some determinants of Latin American markets development. After describing metrics for the financial markets development, they classify the different markets into clusters based on a two-dimension analysis: the depth of the market and the per capita income. They conclude that Latin American markets are less developed than Asian markets and are also unfavorably compared to the countries in the Eastern Europe. The local cluster analysis showed four groups which were studied based on different indicators. The authors highlight that they did not seek causality relations in their study, but there are clear indications of the existence of a virtuous cycle with a direct relation between financial depth and efficiency of financial system.

Claessens, Djankov and Nenova (2000) examine the different financial risk patterns in companies based on a sample of 46 countries and 11,000 companies in the period from 1995

to 1996, by using 12 financial indicators. They also use three performance indicators and come to the conclusion that there are great differences between the countries when it comes to risk and performance of the companies, and that such differences are related to the institutional environment of each country, mostly derived from their legal system. However, besides the legal system, common law or code law, they also indicate the relation between these aspects and the architecture of the financial system with the companies' level of risk.

Caprio and Honohan (2004) study the effect of different market discipline instruments in non-developed markets. The conclusion of this study, which was based on 50 developing countries, indicates that there is no direct relation between low income level and lack of market discipline. The analysis shows that, in function of the existence of foreign banks in countries which have low income level, the discipline is put into practice following the standards of such banks' headquarters. According to the authors, there is more financial discipline in these countries as previously thought. They suggest that the countries, when trying to have more discipline, reduce the participation of the State in banks, eliminate deposit guarantees and do not completely trust the banking supervision as a reason for discipline. More transparency is one of their suggestions to improve discipline and market development.

Barth, Caprio and Levine (2000) investigate bank regulation and ownership in 60 countries making a relation between such indicators and market stability and development. Their main conclusions are: there is no direct relation between restrictions imposed by regulation and level of financial market development; there is no relation between bank ownership (or restrictions to it) and financial market development. They also conclude that it is more probable that countries with higher levels of restriction for banks to act in the securities market will face a crisis in the banking system, i.e., more regulation restrictions cause more instability in the system. The authors argue that not allowing banks to own non-financial companies is harmful to the stability of the financial system. Evidence shows that more integration between banks and other activities is a positive thing to market development. Finally, the empirical evidence indicates that bank ownership by the State is harmful to market development, either regarding the banking system or the stock market. The research shows the negative implications of imposing more restrictions to the banking activity.

Table 1 presents an overview of the works described in this literature review:

Table 1 - Relevant studies and variables

Author	Date	Study	Relevant Variables
La Porta <i>et al</i>	1998 1999	Economic development	Origin of legal system Protection to creditor
Dermigue-Kunt and Levine	1999	Financial markets	Origin of legal system Other cultural / legal characteristics
La Porta <i>et al</i>	2000	Financial markets	Efficiency of legal system
Beck and Levine	2003	Financial markets	Protection to private property Origin of legal system
Tadesse	2002	Economic development	Financial architecture
Chinn and Ito	2005	Stock market	Financial liberalization
Levine and Zervos	1996	Economic development	Stock market International integration
Alfaro <i>et al</i>	2004	Economic development	Foreign direct investment
Beck <i>et al</i>	2000	Economic development	Financial structure Protection to investors
Ndikumana	2001	Domestic investment	Financial markets development
Francis <i>et al</i>	2001	Accounting systems	Origin of legal system
Fuchs-Schündeln and Funke	2001	Economic development	Liberalization of markets
Boutchkova and Megginson	2000	Stock markets	Privatizations
Modigliani and Perroti	1997	Financial markets	Protection to investors
Herrero <i>et al</i>	2002	Financial markets	Markets' depth Per capita income
Claessens <i>et al</i>	2000	Companies' risk	Origin of legal system
Caprio and Honohan	2004	Market discipline	Level of income
Barth <i>et al</i>	2000	Market development and stability	Regulation Bank ownership

Source: own elaboration.

We have thus presented a panorama of the main studies regarding financial markets development. Our next step is analyzing the determinants of market development based on what has already been studied and with the inclusion of new variables.

III – Data

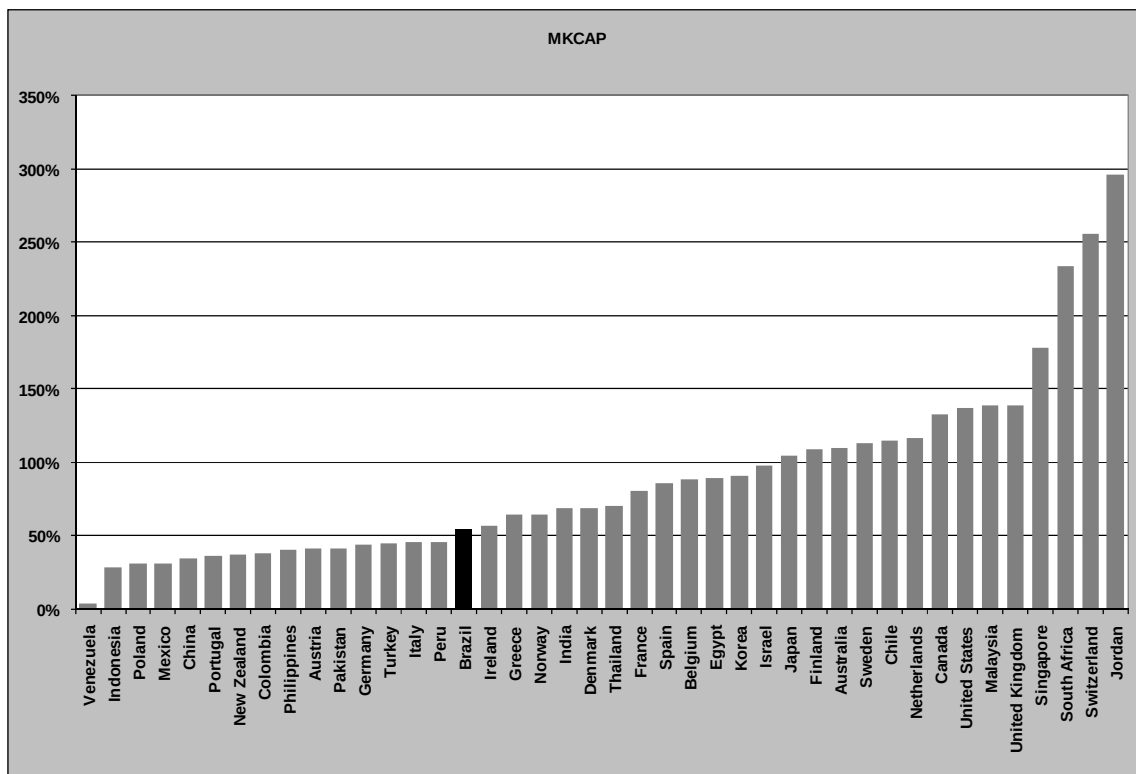
We will analyze this development from three complementary points of view. The stock market development is measured by the quotient between the capitalization value of companies listed in each country and its GDP (MKCAP), according to what was proposed by Tadesse (2002); and the bank market development is measure by the quotient between the long term debt volume and the GDP (LTD), also according to Tadesse (2002). The source of

data is the World Bank Annual Report. A last dependent variable indicates the market orientation of each country: whether it is more oriented to stock market or to credit market. We will use a variable also developed by Tadesse (2002), who called it Architecture of the Market (ARQ), which has been cited in the literature review. We obtained data from 42 countries which are listed in Table 2, as follows:

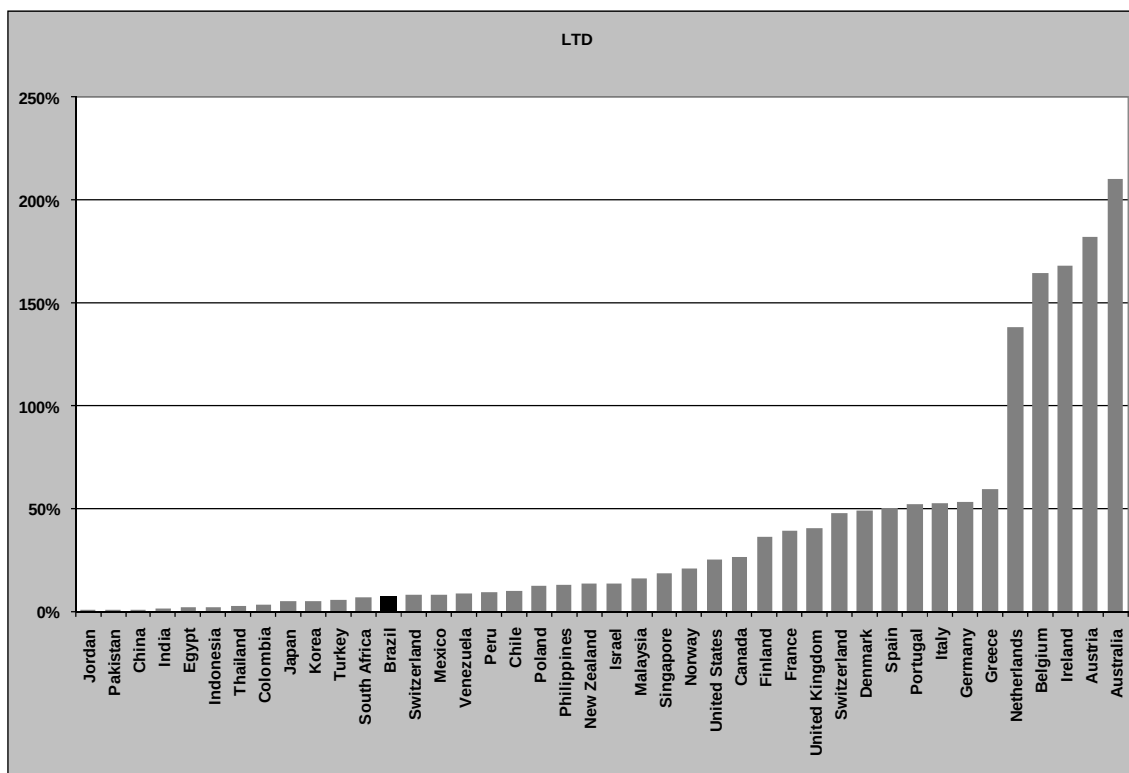
Table 2 - Countries in the sample

Australia	Colombia	India	Korea	Peru	Sweden
Austria	Denmark	Indonesia	Malaysia	Philippines	Switzerland
Belgium	Egypt	Ireland	Mexico	Poland	Thailand
Brazil	Finland	Israel	Netherlands	Portugal	Turkey
Canada	France	Italy	New Zealand	Singapore	United Kingdom
Chile	Germany	Japan	Norway	South Africa	United States
China	Greece	Jordan	Pakistan	Spain	Venezuela

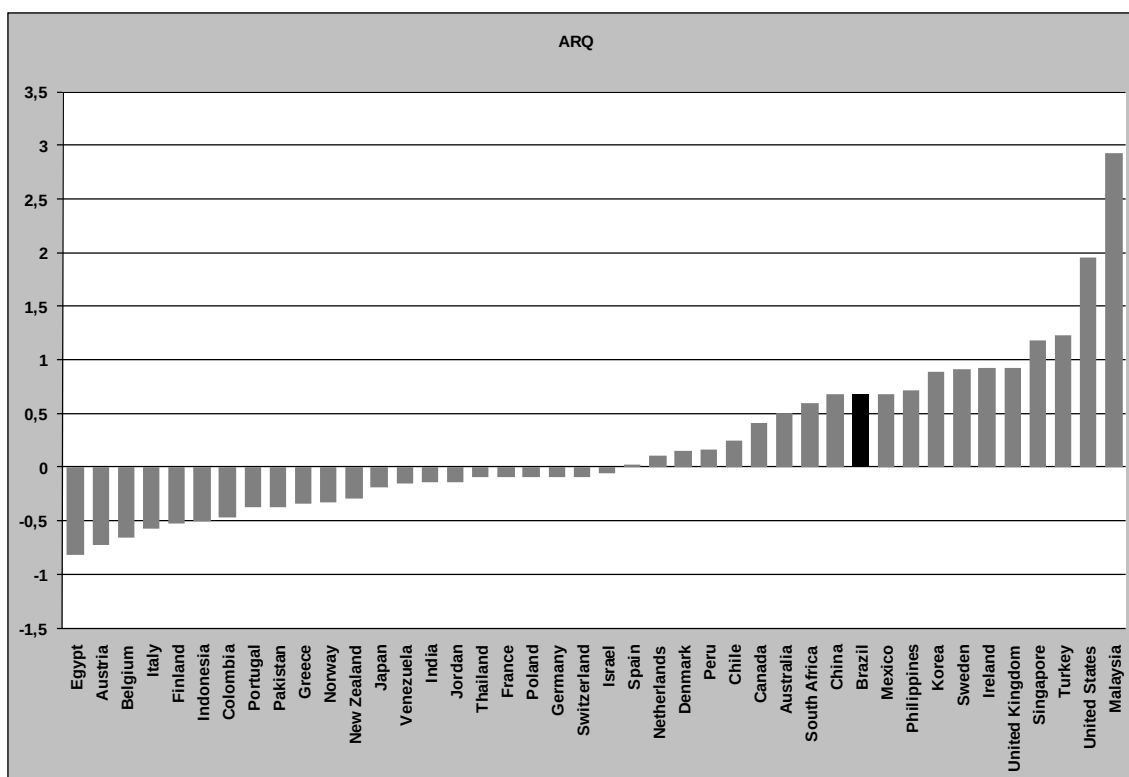
The following graphs show the three dependent variables of this study.



Graph 1 - Value of companies listed on the stock exchange



Graph 2 - Total value of long term credits



Graph 3 - Variable Architecture of the Markets

Brazil holds the 27th place in stock market development and the 30th in credit market development. India holds the 23rd and 39th places and China holds the 38th and 40th, both respectively. There is an indication that the three countries are distant from the most developed countries regarding such two aspects. When the analysis is carried out through variable ARQ, Brazil takes the 12th place, China takes the 15th place and India takes the 27th place among the countries in the sample. This fact shows that Brazil and China would be among the countries in which the economy is more oriented to the stock market.

There are 17 determinants, or independent variables, in the model. They may be classified as related to general economic conditions of the studied country, its level of international market insertion, or globalization, and cultural. Table 3 presents the variables, their sources, period of data analysis, metrics and abbreviation, classification and indication of relation.

Table 3 - Variables of the study: origin, period, classification and indication

Variable	Description	Source	Period	Classification	Indication
MKCAP	Value of the companies listed in the country / GDP	World Bank	2005		
ARQ	Architecture of the financial market	Tadesse (2002)	2000		
LTD	Long term debt / GDP	World Bank	2005		
SUP	Surplus or deficit in public accounts / GDP	World Bank	2005	Economy	Equilibrium in public accounts
EXPOR	Exports / GDP	World Bank	2005	Internationalization	Level of globalization
IDE	Foreign direct investment / GDP	World Bank	2005	Internationalization	Level of globalization
GNIC	Per capita income / Total of per capita income for the sample	World Bank	2005	Economy	Individual wealth
GCF	Gross capital formation / GDP	World Bank	2005	Economy	Capital formation
IMP	Imports / GDP	World Bank	2005	Internationalization	Level of globalization
IVA	Value aggregated by the industry / GDP	World Bank	2005	Economy	Sophistication of the industrial park
INF	Annual inflation	World Bank	2005	Economy	Stability
POP	Population growth	World Bank	2005	Economy	Stability
UNIV	% of undergraduate students	World Bank	2005	Culture	Presence in universities
DAYS	Time to open a business (days)	World Bank	2005	Culture	Entrepreneurship
LI	Press freedom	Reportes Sans Frontieres	2005	Culture	Freedom
CORR	Corruption	Transparencia Internacional	2005	Culture	Corruption
DI	Information Diffusion	Bushman <i>et al</i> (2004)	2003	Culture	Information
LEC	Economic freedom	Heritage Foundation - Wall Street Journal	2005	Culture	Entrepreneurship
OSL	Origin of legal system	Beck <i>et al</i> (2000)	1999	Culture	Legal system

We observed that, in the three cases, the periods of data analysis are different. They are the variables ARC, DI and OSL. The last one is naturally timeless; and we can suppose that the other two have a very slow speed of modification, since they are much more associated to the perennial characteristics of economies. That permits using data from different periods in the analysis without affecting the robustness of the results.

After adding the 15 listed variables to the two pieces of information about financial markets, there is 966 possible data (42 countries x 23 pieces of information). We did not get 11 pieces of information and six of them are about the variable DI (Information Diffusion).

The missing data were substituted by the means of the respective variables, according to what was proposed by Gujarati (2003).

Table 4 presents the statistics of the different variables.

Table 4: Variables – Descriptive Statistics

	MKCAP	ARC	LTD	SUP	EXPOR	IDE	GNIC	GCF	IMP	IVA	INF	POP	UNIV	DAYS	LI	CORR	DI	LEC	OSL
Mean	0.88	0.25	0.38	-0.01	0.44	0.03	0.02	0.22	0.42	0.32	0.04	0.01	0.52	36.33	17.80	6.06	73.11	68.28	0.60
Median	0.69	0.07	0.14	0.00	0.37	0.03	0.02	0.21	0.33	0.30	0.03	0.01	0.56	25.50	8.50	6.40	77.15	67.55	1.00
Maximum	2.96	2.93	2.10	0.16	2.44	0.17	0.07	0.43	2.15	0.52	0.29	0.02	0.92	152.00	83.00	9.60	96.72	85.70	1.00
Minimum	0.03	-1.10	0.00	-0.16	0.10	-0.15	0.00	0.15	0.11	0.21	-0.02	0.00	0.05	2.00	0.50	2.10	29.51	47.70	0.00
Standard Deviation	0.62	0.81	0.54	0.05	0.39	0.04	0.02	0.05	0.34	0.08	0.05	0.01	0.23	37.04	19.80	2.51	-0.50	9.69	0.50
Skewness	1.58	1.15	2.00	0.42	3.64	-0.47	0.34	2.01	3.37	1.00	3.33	0.37	-0.12	2.06	1.41	-0.07	-0.50	-0.05	-0.39
Kurtosis	5.49	4.63	5.91	8.30	18.67	9.37	1.84	8.08	16.79	3.11	17.01	2.28	2.10	6.66	4.40	1.52	2.20	2.01	1.15
Jarque-Bera	28.30	13.9	42.72	50.36	522.74	72.54	3.16	73.43	412.47	6.98	421.12	1.84	1.51	52.97	17.28	3.88	2.88	1.72	7.04
Prob.	0%	0%	0%	0%	0%	0%	21%	0%	0%	3%	0%	40%	47%	0%	0%	14%	24%	42%	3%
Number of Observations	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42	42

Some details can be noticed in the analysis of such statistics. Firstly, the greater volume in company value in relation to debt; although the company value incorporates the debt value, there is still an equity value (difference between MKCAP and LTD) superior to that of indebtedness, what may suggest the low use of leverage by the companies.

There is another interesting feature concerning imports and exports. The mean values are very similar and that indicates equilibrium in the analyzed countries. The analysis of minimum and maximum of other data, such as DAYS – number of necessary days to open a business, CORR – corruption, LEC – economic freedom, shows the great diversity between the countries.

The analysis of correlations between the variables, which is presented below, indicates the existence of multicollinearity.

Table 5 – Variables - Correlations

	EXPOR	IDE	GNIC	GCF	IMP	IVA	INF	POP	UNIV	DAYS	LI	CORR	DI	LEC	OSL
SUP	19%	0%	40%	-2%	7%	34%	11%	-10%	33%	-2%	-22%	42%	34%	37%	1%
EXPOR		42%	10%	-6%	96%	26%	-14%	30%	8%	-20%	17%	27%	15%	33%	-22%
IDE			-17%	-19%	50%	-3%	-8%	10%	-13%	-8%	29%	6%	-6%	-1%	5%
GNIC				-19%	3%	-35%	-41%	-47%	72%	-51%	-68%	86%	90%	74%	18%
GCF					-4%	27%	-2%	-6%	-6%	0%	26%	-18%	-22%	-16%	-33%
IMP						14%	-22%	34%	6%	-26%	19%	24%	10%	31%	-27%
IVA							52%	27%	-28%	37%	36%	-31%	-29%	-30%	-20%
INF								37%	-38%	65%	26%	-50%	-40%	-54%	4%
POP									-50%	21%	51%	-36%	-49%	-21%	-41%
UNIV										-50%	-67%	71%	81%	60%	15%
DAYS											23%	-60%	-52%	-59%	11%
LI												-64%	-71%	-54%	-20%
CORR													88%	90%	5%
DI														76%	11%
LEC															-16%

According to Gujarati (2003), who suggests the existence of multicollinearity when the correlation between the variables is equal or superior to 80%, we noticed indications of the existence of multicollinearity between the variables GNIC and CORR and DI, as well as between UNIV and DI and EXP and IMP. Such correlations are highlighted in gray.

Besides correlation, we use another measure to identify multicollinearity: VIF - Variance Inflation Factor. Table 6 shows the results for VIF and for Tolerance (1/VIF). The gray cells indicate variables which have problems of multicollinearity. They coincide with those indicated by the correlation analysis, excepting variable UNIV, which did not present any problem in this analysis.

Table 6: Tolerance and VIF

	MKCAP	LTD	SUP	EXP	IDE	GNIC	GCF	IMP	IVA	INF	POP	UNIV	DAYS	LI	CORR	DI	LEC	OSL
Tolerance	0.33	0.47	0.38	0.02	0.26	0.09	0.49	0.02	0.25	0.30	0.28	0.19	0.35	0.20	0.06	0.07	0.12	0.44
VIF	3.00	2.12	2.64	42.45	3.92	11.27	2.05	47.17	3.97	3.35	3.58	5.27	2.85	4.92	16.43	14.24	8.50	2.30

When modeling, we avoided including these variables together with those with which they presented multicollinearity.

IV – Methodology and Results

The influence of different variables on financial markets development will be estimated through the following multiple regression:

$$D_i = \alpha_i + \beta_{1i}V_1 + \beta_{2i}V_2 + \dots + \beta_{14i}V_{14} + \varepsilon$$

in which D_i represents the three dependent variables that will be used in the study (ARQ, MKCAP, LTD) and V_{ni} represent the different explicative variables. The correction to heterocedasticity was done according to the processes proposed by White and constant in Gujarati (2003).

We examined several possible configurations of the models, with different transformations of the variables, and selected the three best of each of the dependent variables through the Akaike and Schwarz's criteria.

We obtained the following model for the dependent variable MKCAP:

Table 7: Model MKCAP

Variable	Coefficient	Std. Error	t-Statistic	Prob.	Sig.
IMP	0.449	0.254	1.769	8.53%	*
IDE	4.916	2.275	2.161	3.74%	**
EXP(GNIC)	0.514	0.153	3.368	0.18%	***
1/POP	5.6E-05	2.1E-05	2.656	1.17%	**
(LI)^2	-1.2E-04	4.7E-05	-2.526	1.61%	**
1/DAYS	1.296	0.476	2.725	0.99%	***
Significance:	* 10%	** 5%	*** 1%		

It indicates that the greater stock market development is associated to more globalization, either through the variable IMP or IDE, higher per capita income, lower population growth rate, more press freedom, in an accelerated way as showed by the transformation of the variable and, finally, more easiness to develop a business. One should notice that the coefficients for POP and LI are small. The adjusted R2 in this model is 33.63%. The signs were as expected, which show the influence of different factors on stock markets development.

When the dependent variable is long term debt, LTD, the obtained model was as follows:

Table 7: Model LTD

Variable	Coefficient	Std. Error	t-Statistic	Prob.	Sig.
IMP	0.493	0.152	3.245	0.25%	***
IDE	-4.740	1.792	-2.645	1.20%	**
EXP(GNIC)	4.886	2.556	1.911	6.39%	*
1/POP	-3.82E-05	1.82E-05	-2.094	4.34%	**
LI	-0.007	0.003	-2.126	4.05%	**
EXP(INF^2)	-4.549	2.577	-1.765	8.61%	*
Significance:	* 10%	** 5%	*** 1%		

Two of the signs in this model are the same as those obtained from the model regarding the stock market. The better the country's level of globalization (given by the variable IMP), the greater the credit market development. The same occurs with variable GNIC which represents per capita income. The inflation indicator presents an interesting result. The size of the credit market is sensitive to high inflation rates, but not to low rates. Thus, countries whose inflation rate is high do not have a developed credit market. That is also an expected result. The result regarding population is as well interesting: larger credit markets are associated to higher growth rates. Nevertheless, one must take into account that the coefficient is very low and that fact shows a restricted influence of this indicator. Maybe that is a contraposition to the lower stock market development as a source of revenue in the

companies. The coefficient of variable IDE shows that in countries which receive less foreign capital there is more development in the credit market, and that indicates a reaction of the economies to the lack of an important alternative for companies' financing. Finally, variable LI presents a negative sign; that indicates that in countries which have less press freedom there is more development in the credit or bank market. It is possible no see here a clear contraposition to the result obtained for the stock market, in which the sensitivity to market size is visibly associated to more press freedom.

The model to the dependent variable ARQ – market architecture is as follows:

Table 9: Model ARQ

Variable	Coefficient	Std. Error	t-Statistic	Prob.	Sig.
EXP(EXPOR)	0.114	0.059	1.951	5.91%	*
1/(IDE)	-3.73E-04	1.79E-04	-2.086	4.43%	**
1/DAYS^2	1.350	0.654	2.065	4.64%	**
EXP(DI)	1.58E-42	2.04E-43	7.722	0.00%	***
1/(INF)^2	2.82E-06	1.04E-06	2.696	1.07%	**
1/GNIC^2	-4.11E-07	1.70E-07	-2.414	2.11%	**
SUP	-4.718	2.272	-2.077	4.52%	**
Significance:	* 10%	** 5%	*** 1%		

Variable ARQ shows how much a country is oriented to stock market or credit market. The bigger it is, the more stock market-oriented. Based on that, one can notice that countries with higher level of globalization, given by variable EXPOR, are more stock market-oriented. Greater volume of IDE is also a characteristic of countries with more developed stock market, and so is DI as well. An interesting result concerns variable DAYS. The coefficient associated to transformation $1/DAYS^2$ shows that having an economy focused on entrepreneurship – which is represented here by few necessary days to open a business – is essential to stock markets development. The same result was obtained for variable INF, and that indicates that high inflation rates are prejudicial to stock markets development; that leads the country to be more oriented to credit market. The result for variable GNIC comes as well as no surprise. There is a strong stimulation to financial markets, especially the stock market, caused by high levels of per capita income. Finally, variable SUP is interesting because it shows that stock markets would be more developed in countries with more public deficit. This may be explained by the presence of countries whose stock market is more developed, such as USA and United Kingdom, with high deficit level.

V – Conclusions

It is possible to draw several conclusions from this study, as well as many doubts. Table 10 presents a summary of the obtained results:

Table 10: Summary of the results

MKTCAP		LTD		ARQ	
				EXP(EXPOR)	0.114
IMP	0.449	IMP	0.493		
IDE	4.916	IDE	-4.740	1/(IDE)	-3.73E-04
EXP(GNIC)	0.514	EXP(GNIC)	4.886	1/GNIC^2	-4.11E-07
1/POP	5.6E-05	1/POP	-3.82E-05		
(LI)^2	-1.2E-04	LI	-0.007		
1/DAYS	1.296			1/DAYS^2	1.350
		EXP(INF^2)	-4.549		
				EXP(DI)	1.58E-42
				1/(INF)^2	2.82E-06
				SUP	-4.718

The coefficients highlighted in gray indicate signs opposed to what was found in the models for other dependent variables. We may thus conclude that countries with developed stock markets are more globalized, have more foreign capital, higher per capita income, more press freedom, are more oriented to entrepreneurship and have lower rate of population growth. Many of such affirmations may be repeated regarding the market as a whole. Countries in which it is more developed are as well more globalized, have higher per capita income, are more oriented to entrepreneurship and have low inflation rates. The results regarding the credit market also indicate that countries with more relative credit volume are as well more globalized, have more press freedom, higher per capita income and lower inflation rates. They also present more population growth and lower volume of foreign direct investments. In the first case, there is a direct relation between population growth and legal system with French origin. According to the literature, in countries whose origin is French, there is less stock market development; it is substituted by the credit market as a source of revenue for companies. In the second case, we can explain the coefficient's sign by a substitution effect. There are no external resources; the internal possibilities are somehow developed.

In comparison with previous studies, we find interesting results. In the case of variable IDE, which Alfaro *et al* (2004) argue to have little relevance to analyze market development, we come to a different conclusion. In the three estimated models we got significant coefficients in 5%; and in the models in which the dependent variables were MKCAP and

ARC, there is a clear indication that countries with more developed markets and more oriented to stock market have higher volumes of foreign direct investment. This very same variable is negatively associated to credit market development, and that suggests a contraposition of it in relation to stock markets.

GNIC is as well a variable which was used both in this study and in previous works. Caprio and Honohan (2004) conclude that there is no relation between income level and market discipline. We study other dependent variables and come to the conclusion that per capita income is related to level of market development. Our analysis indicates that there is an exponential relation between level of market development in different countries and their per capita income.

The continuation of the results analysis goes necessarily through market efficiency analysis, since one of the important identified variables is press freedom. Such variables may be considered a proxy for information fluidity, which is fundamental to the development of efficient markets. It is also important to highlight the role of more globalization in this context. Countries which have higher level of globalization tend to have more efficient markets even due to the creation of greater external communicating vessels, what leads them to have more transparency in their actions. It becomes clear in the risk analysis carried by risk classification agencies, in which much weight is given to the so-called international exposure indicator, which is the result of the division of exports plus imports by the total GDP.

Another factor that must be noticed is the entrepreneurship bias. The number of days to open a new business was used as a proxy thereof. There are clear indications that countries in which the institutional environment is more favorable to business have more developed markets, particularly stock markets. This result is really interesting to regulators, since there are ways to alter this pattern in countries in which the environment is not favorable to business.

Per capita income is as well a determinant of the level of market development. We can thus expect the growth of this factor, together with other indicators, to bring more development to the markets.

The next question is: How are Brazil, India and China in relation to the variables which characterize countries with more developed markets? Table 11 presents the position of these three countries in the ranking for different indicators – the first position is always the best one in the rankings.

Table 11 – Brazil, India and China: Indicators

	EXP	IDE	GNIC	IMP	INF	POP	DAYS	LI	DI
Brazil	40	12	32	41	37	28	42	27	36
China	21	32	37	23	25	14	35	42	38
India	37	6	41	34	27	30	38	32	42

The table shows that the three countries are still very far from having the basic characteristics to present good financial market development. All of them are in the end of the list when the 42 countries of the sample are classified by variable DAYS, which indicated the necessary days to open a new business. This case is not very different from DI (information diffusion), LI (press freedom), INF (inflation) and GNIC (per capita income). China presents better results than Brazil and India for the variables IMP (imports) and EXP (exports), which indicate international integration, and POP (population growth). We can anyway conclude that the three countries are still very far from countries with more developed markets, and that a focused effort must be taken so that their position may improve in some years.

It is important to notice that some of the variables selected initially were not significant when the analysis was carried. Some of them were excluded in the beginning of the analysis because they could be replaced by others, given their high correlation. This was the case for the variables CORR (corruption) and DI (information diffusion). The variables IMP (imports) and EXP (exports) were alternately used also due to their high collinearity. There is possibly cases of multicollinearity between LEC (economic freedom), UNIV (% of undergraduate students) and other variables such as GNIC (per capita income), as both presented high correlation of 74%. Among the variables which were not significant in the regressions, the one that surprised the most was OSL (origin of legal system). In other studies, such as Dermigue-Kunt and Levine (1999), this variable was significant in relation to financial markets development. It did not happen in our work. Even in a simple regression the results were not significant; sometimes significant, but with negative or null adjusted R^2 . We may argue that this construction was not the most adequate, as it deals with a dummy variable among many others with characteristics that makes them approximate to continuous variables. We may as well imagine multicollinearity effects that the analysis has not captured. Then we may have an indication of the origin of legal system being hidden in one of the variables. It was anyway remarkable that we found no significant relation between origin of legal system and markets development.

In this study we analyzed the characteristics of countries and the financial markets' level of development. We built a panorama of the moment and compared the financial markets development based on different indicators. A natural way for further studies is elaborating a panel study with the purpose of analyzing the evolution of the characteristics and markets. Moreover, it is important to obtain data from Russia, so that one may have a complete panel for BRICs. Nevertheless, it is important to emphasize that it is difficult to gather long term data for most of the analyzed countries. Rebuilding some variables and including others is as well a way to continue this work.

References

- Alfaro, L.A., Chanda, S., Kalemli-Ozcan, Sayek, S., 2003. FDI and economic growth: The role of local financial markets. *Journal of International Economics*, 64, 89–112.
- Barth, J.R., Caprio, C., Levine, R., 2000. Banking systems around the globe: Do regulations and ownership affect performance and stability? World Bank Policy Research Working Paper n. 2325, [ssrn-id 630697](#).
- Beck, T., Levine, R., 2003. Legal institutions and financial development. NBER Working Paper n° 10126, [ssrn-id636556](#).
- Beck, T., Levine, R., Demirgüç-Kunt, Maksimovic, V., 2000. Financial structure and economic development: Firm, industry, and country evidence. World Bank Policy Research Working Paper n. 2423.
- Boutchkova, M.K., Megginson, W.L., 2000. Privatization and the rise of global capital markets. *Financial Management*, 29(4), 31-75.
- Caprio, C., Honohan, P., 2004. Can the unsophisticated market provide discipline. World Bank Policy Research Working Paper n. 3364, [ssrn-id 625241](#).
- Chinn, M.D., Ito, H., 2005. What matters for financial development? Capital controls, institutions and interactions. Working Paper University of Wisconsin and Portland State University, [ssrn-id677506.pdf](#).
- Claessens, S., Djankov, S., Nenova, T., 2000. Corporate risk around the world. World Bank Policy Research Working Paper n. 2271, [ssrn-id629121](#).
- Dermigue-Kunt, A., Levine, R., 1999. Bank based and market based financial systems: Cross-country comparisons. World Bank Working Paper, [ssrn-id569255](#).
- Fan, J.P.H., Titman, S., 2003. An international comparison of capital structure and debt maturity choices. Working Paper Hong Kong University of Science and University of Texas at Austin, [ssrn-id423483](#).
- Francis, J.R., Khurana, I.K., Pereira, R., 2001. Investor protection laws, accounting and auditing around the world. Working Paper University of Missouri, [ssrn-id287652](#).
- Fuchs-Schündeln, N., Funke, N., 2001. Stock Market liberalizations: Financial and macroeconomic implications. IMF Working Paper WP/01/193, [ssrn-id296509](#).
- Goldsmith, R.W., 1969. *Financial Structure and Development*. New Haven, CT, Yale University Press.
- Gujarati, D., 2003. *Basic Econometrics*. fourth ed. McGraw-Hill, Irwin.
- Herrero, A.G., Santillan, J., Gallego, S., Cuadro, L., Egea, C., 2002. Latin American financial development in perspective. Madrid Seminar of the Eurosystem and Latin American Central Banks. Madrid.

Korajczyk, R.A., 1996. A measure of stock market integration. *World Bank Economic Review*, 10(2), 267-289.

La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R.W., 1998. Law and finance. *Journal of Political Economy*, 52, 1131-1150.

La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R.W., 1999. The quality of government. *Journal of Law and Economics Organization*, 15, 222-279.

La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R.W., 2000. Investor protection and corporate governance. *Journal of Financial Economics*, 58, 3-27.

Levine, R., Zervos, S., 1996. Stock market development and long-run growth. The World Bank, Policy Research Working Paper 1582, [ssrn-id614946](#).

Modigliani, F., Perotti, E., 1997. Protections of minority interest and the development of security markets. *Managerial and Decision Economics*, 18, (7/8), 519-528.

Ndikumana, L., 2001. Financial development, financial structure and domestic investment: International evidence. Working Paper n° 16 PERI - University of Massachusetts.

Tadesse, S., 2002. Financial architecture and economic performance: International Evidence. Working Paper, The University of South Carolina, [ssrn-id307223.pdf](#).

World Bank – World Development Indicators, 2005.

Appendix 1: Data

Pais	MKCAP	LTD	ARQ	SUP	EXP	IDE	GNIC	GCF	IMP	IVA	INF	POP	UNIV	DAYS	LI	CORR	DI	LEC	OSL
Australia	110%	210%	2.10	1.2%	18.4%	-4.7%	3.7%	25.7%	21.2%	27.1%	4.7%	1.2%	71.8%	2	6.5	8.8	89.25	82.7	0
Austria	41%	182%	1.82	-2.0%	53.2%	3.0%	4.2%	21.2%	47.8%	30.7%	2.0%	0.7%	50.3%	29	2.5	8.7	87.53	71.3	1
Belgium	88%	164%	1.64	-0.1%	87.1%	8.6%	4.0%	21.4%	84.8%	24.0%	2.2%	0.6%	63.0%	34	4.0	7.4	96.73	74.5	1
Brazil	54%	8%	0.08	0.0%	15.1%	1.7%	0.4%	16.0%	11.5%	30.3%	7.5%	1.4%	23.8%	152	14.5	3.7	56.14	60.9	1
Canada	133%	26%	0.26	1.7%	38.6%	3.1%	3.7%	21.0%	34.4%	31.4%	3.2%	1.0%	62.3%	3	4.5	8.4	93.37	78.7	0
Chile	115%	10%	0.10	4.6%	41.0%	5.6%	0.7%	22.4%	32.5%	46.8%	7.9%	1.1%	47.8%	27	11.8	7.3	62.46	78.3	1
China	35%	0%	0.00	-2.1%	37.3%	3.5%	0.2%	43.3%	31.7%	47.3%	4.4%	0.6%	20.3%	48	83.0	3.2	50	54	0
Colombia	37%	3%	0.03	3.9%	21.4%	8.4%	0.3%	20.4%	22.2%	34.4%	5.7%	1.4%	29.3%	43	40.2	4.0	58.13	60.5	1
Denmark	69%	49%	0.49	1.8%	48.7%	2.0%	5.4%	20.9%	43.9%	24.7%	2.6%	0.3%	80.5%	5	0.5	9.5	95.52	77.6	1
Egypt	89%	2%	0.02	-5.8%	30.3%	6.0%	0.1%	18.0%	32.6%	36.1%	6.2%	1.9%	33.9%	22	52.0	3.4	60	53.2	1
Finland	108%	36%	0.36	3.2%	38.7%	2.1%	4.2%	20.2%	35.2%	29.6%	1.6%	0.3%	91.7%	14	0.5	9.6	94.82	76.5	1
France	80%	39%	0.39	-2.8%	26.1%	3.3%	3.9%	20.2%	27.1%	20.9%	1.9%	0.6%	56.4%	8	6.3	7.5	86.14	66.1	1
Germany	44%	54%	0.54	-2.3%	40.1%	1.1%	3.9%	17.2%	35.1%	29.7%	0.5%	-0.1%	60.0%	24	4.0	8.2	90.99	73.5	1
Greece	64%	59%	0.59	-5.1%	20.8%	0.3%	2.2%	23.8%	28.0%	20.8%	3.7%	0.4%	89.1%	38	4.0	4.3	72.07	57.6	1
India	69%	1%	0.01	-3.6%	20.3%	0.8%	0.1%	33.4%	23.3%	27.3%	4.5%	1.4%	11.4%	71	27.0	2.9	29.51	55.6	0
Indonesia	28%	2%	0.02	-1.1%	32.9%	1.8%	0.1%	21.8%	28.6%	44.9%	14.8%	1.4%	17.1%	151	26.0	2.2	60	55.1	0
Ireland	57%	168%	1.68	1.0%	83.2%	-14.7%	4.6%	25.0%	67.8%	37.3%	3.5%	2.2%	59.3%	24	0.5	7.4	83.34	81.3	0
Israel	97%	14%	0.14	-2.6%	45.9%	4.5%	2.1%	18.5%	51.3%	30.0%	0.6%	1.8%	57.9%	34	10.0	6.3	82.47	68.4	0
Italy	45%	52%	0.52	-3.5%	26.3%	1.1%	3.4%	20.9%	26.4%	26.9%	2.1%	0.7%	65.6%	13	8.7	5.0	78.98	63.4	1
Japan	104%	5%	0.05	0.0%	13.4%	0.1%	4.4%	22.7%	11.4%	30.2%	-1.8%	0.0%	55.3%	31	8.0	7.3	91.79	73.6	1
Jordan	296%	0%	0.00	-4.7%	51.9%	12.1%	0.3%	23.7%	93.4%	29.6%	4.0%	2.3%	39.2%	18	24.0	5.7	60	64	0
Korea	91%	5%	0.05	0.8%	42.3%	0.5%	1.8%	30.1%	39.9%	40.3%	-0.2%	0.4%	89.9%	22	7.5	5.0	83.5	68.6	0
Malaysia	139%	16%	0.16	-4.3%	123.4%	3.0%	0.6%	19.9%	99.9%	51.8%	4.6%	1.8%	32.0%	30	33.0	5.1	63.83	65.8	0
Mexico	31%	8%	0.08	-1.2%	30.0%	2.4%	0.8%	21.8%	31.5%	26.0%	5.5%	1.0%	24.0%	58	45.5	3.5	59.95	65.8	1
Netherlands	117%	138%	1.38	0.0%	71.2%	6.5%	4.4%	19.2%	63.0%	24.4%	1.6%	0.2%	60.7%	11	0.5	8.6	92	77.1	1
New Zealand	37%	13%	0.13	4.6%	29.0%	1.8%	2.9%	24.6%	30.1%	24.9%	2.5%	0.9%	81.9%	12	2.0	9.6	90	81.6	0
Norway	65%	21%	0.21	16.3%	45.3%	1.1%	6.8%	20.6%	28.1%	43.4%	8.4%	0.7%	79.7%	13	0.5	8.9	95.31	70.1	1
Pakistan	41%	0%	0.00	-3.2%	15.5%	2.0%	0.1%	18.1%	19.3%	26.5%	8.7%	2.4%	4.7%	24	60.8	2.1	32.47	58.2	0
Peru	45%	9%	0.09	-0.8%	24.8%	3.2%	0.3%	18.6%	19.2%	34.8%	3.4%	1.5%	33.5%	102	33.3	3.5	40.33	62.1	1
Philippines	41%	13%	0.13	-3.0%	47.3%	1.2%	0.1%	15.1%	52.0%	32.3%	6.2%	1.8%	28.1%	48	50.0	2.5	44.26	57.4	1
Poland	31%	13%	0.13	-2.3%	37.2%	3.2%	0.8%	19.2%	37.5%	30.8%	2.6%	0.0%	63.4%	31	12.5	3.4	60	58.8	1
Portugal	37%	52%	0.52	-5.8%	28.5%	1.7%	1.9%	22.3%	37.4%	24.7%	2.7%	0.5%	56.5%	54	4.8	6.5	70.59	66.7	1
Singapore	179%	18%	0.18	4.1%	244.3%	17.2%	3.0%	19.0%	214.8%	33.8%	0.4%	2.4%	60.0%	6	50.7	9.4	83.72	85.7	0
South Africa	234%	7%	0.07	0.2%	26.8%	2.6%	0.5%	18.0%	28.3%	30.3%	4.8%	1.1%	15.3%	35	6.5	4.5	59.56	64.1	0
Spain	85%	50%	0.50	1.5%	25.4%	2.0%	2.8%	29.7%	30.6%	29.5%	4.4%	1.7%	66.9%	47	8.3	7.0	75.31	70.9	1
Sweden	113%	48%	0.48	2.1%	48.6%	3.0%	4.6%	17.1%	40.9%	28.2%	1.2%	0.4%	81.7%	16	2.0	9.2	95.47	72.6	1
Switzerland	256%	8%	0.08	0.6%	46.2%	4.2%	6.2%	20.4%	38.9%	28.3%	0.6%	0.6%	47.4%	20	0.5	9.1	93.78	79.1	1
Thailand	70%	3%	0.03	2.5%	73.8%	2.6%	0.3%	31.7%	75.4%	44.2%	4.5%	0.8%	43.0%	33	28.0	3.8	52.55	65.6	0
Turkey	44%	5%	0.05	-15.5%	27.4%	2.7%	0.5%	24.8%	34.0%	23.7%	5.4%	1.3%	31.2%	9	25.0	3.5	58.55	59.3	1
United Kingdom	139%	41%	0.41	-2.9%	26.1%	7.2%	4.2%	16.8%	30.0%	26.2%	2.0%	0.7%	59.7%	18	5.2	8.6	90.81	81.6	0
United States	137%	25%	0.25	-2.9%	10.1%	0.9%	4.9%	19.2%	15.4%	22.0%	3.0%	1.0%	82.7%	5	9.5	7.6	96.72	82	0
Venezuela	3%	9%	0.09	2.2%	39.7%	2.0%	0.6%	21.0%	17.4%	51.6%	29.0%	1.7%	41.2%	141	23.0	2.3	62.64	47.7	1